4. Optional Activities

1. Area : Food

1.1 Kitchen Gardening

To see how a kitchen garden is planned.

Kitchen Garden :

When vegetables are grown in the space around one's house, it is called a kitchen garden. Plants like coriander, radishes, chillies, fenugreek, brinjal, bitter gourds and various other gourds, okra/lady fingers, cluster beans, ginger are grown in a kitchen garden. It is planned according to the space available. Different types of soil beds are prepared. Flat bed, raised bed and ridged or furrowed bed are the types of beds that can be used.

My Activity

(1) Grow seedlings in empty boxes, coconut shells, paper cups, etc.

(2) Make a list of the vegetables you like.

- Give information in the class about the different types of beds that have to be prepared for growing vegetables.

My Activity

Let's play a game.

Find the names of the vegetables hidden in the box below:

r	x	g	i	n	g	e	r
c	a	z	m	l	q	c	h
a	s	d	f	j	t	s	e
b	b	r	i	n	j	a	l
b	e	a	n	s	c	g	o
a	h	j	u	v	h	a	o
g	p	e	a	s	i	r	n
e	w	k	v	u	l	l	i
x	d	x	f	g	l	i	o
g	o	u	r	d	i	c	n

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Picture reading

- Arrange a visit to a kitchen garden in the neighbourhood and explain how it is laid out.

1.2 Growing Plants in Pots

(Pot Culture)

Obtaining preliminary information about growing plants in pots.

Plants must be chosen according to the size of the pots. Lilies, dahlias, marigolds, asters, roses, hibiscus, *mogara*, etc. can be grown in pots.

First, the pots must be filled with soil mixed with fertilizers.

A fertilizer is a mixture which provides the micronutrients required for the growth of plants. Fertilizers are of two kinds:

1. Organic fertilizers
2. Chemical fertilizers

Watering potted plants

A plant must be watered immediately after it is potted. Then the pot must be placed in the shade. Once the plant has taken root and looks fresh, it can be placed in the sun.

Water the plants in the pots using a watering can with a rose. The amount of water to be given depends on the size of the pot. Too much water will causes the roots to swell. Water the plant when the soil in the pot becomes dry.

Either clay or cement pots can be used for planting. Clay pots are the most suitable for the growth of plants. There should be a hole in the bottom of the pot. This helps to drain the extra water from the soil and to aerate the roots. Pots must be cleaned from the outside at least once a year.

- Give information about pots on a visit to a nearby kitchen garden.

1.3 Fruit Processing

Fruit Trees

Fruit trees can be of different sizes. Mango, jackfruit, *jamun* trees are very big whereas banana, *papaya, chickoo* trees are of medium size. Grapes grow on a climber.

A banana tree has a brittle, soft and thick main stem. It is green in colour. The leaf is broad and very long. First, an inflorescence appears on the banana tree. This develops gradually into fruits i.e. a bunch of bananas.

Banana

The mango is everyone's favourite fruit. The trunk of the mango tree is thick hard and blackish brown. The leaves are deep green in colour. Mango blossoms appear in a particular season. A blossom develops into a green raw mango which in turn becomes a ripe mango. Mangoes are used to make several different foods.

Mango

The trunk of the *chickoo* tree is strong and brown in colour. The leaves are lighter green and of medium size. Flowers are white and the fruit is dark brown and oblong or round. There are black seeds at the centre of the fruit. *Chickoo* is used for eating or for making several food preparations.

Chickoo

My Activity

Draw pictures of the mango, *chickoo* and banana and colour them.

- Visit an orchard in the neighbourhood to get the students to observe the fruit trees and obtain information about them. All kinds of fruits should be eaten when ripe and should be washed well before eating.

1.4 The Fish Industry

Obtaining preliminary information about fishing

What is fishing?

Fish are used by humans as a nutritious food.

Big fish live in seas, rivers, big lakes. Small fish live in streams, ponds, canals, etc. Catching fish is called fishing.

How are fish caught?

There are several methods of catching fish. Very large fish live in seas and big rivers. Mechanical devices and various kinds of nets are used to catch them. We will learn about them next year.

Let us learn about the basic simple methods of fishing.

(1) Standing in the water to catch fish.

Small fish can be caught in this way by standing in shallow water. If the water is deeper, some people may even dive under it to catch the fish. People use their hands to catch small fish.

(2) Using spears to catch fish.

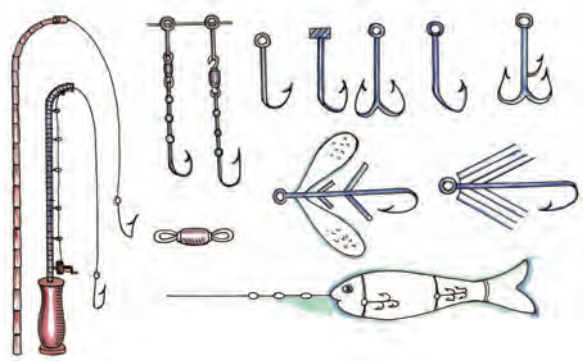
A fish is speared using a hand held spear.

(3) Using a hook to catch fish.

The hook is a very simple and useful fishing device.

It has the shape of the letter J. Its one end is very sharp. A 'bait' or a piece of fish food is tied to it. A string or

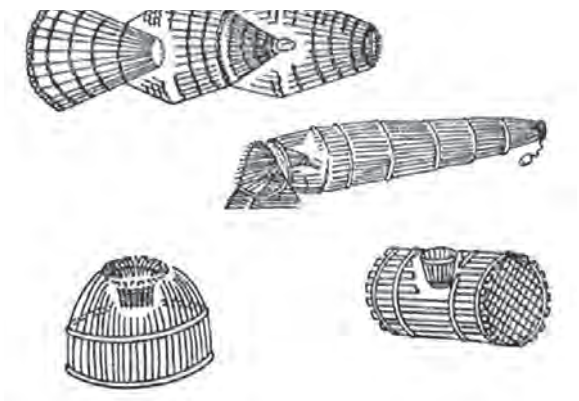
'line' is tied to its other end. A 'float' is tied to the upper end of the string. When the hook is let into the water the fish tries to eat the bait attached to it and the float shakes. The fisherman then pulls the rod out with a jerk. The sharp end of the hook is fixed in the fish's mouth. It is detached and the fish is put in a basket. Another bait is fixed on the hook and lowered into the water. This is how a hook is used for fishing.



Fish Hooks

(4) Using traps for fishing.

Thin strips of bamboo are used to make a trap. A trap is shaped like a *'mrudang'*. The bait is placed inside the trap. Sharp strips of bamboo are fixed on the inner side of the trap. As a result, once the fish enters the trap, it cannot go out again. They get trapped inside.



Traps made from thin strips of bamboo

These are four simple and basic methods of fishing. People who live on river banks know about these methods.

My Activity

(1) Make a model of a fish hook.

(2) Draw a picture of a fish trap.

(3) If possible, go with some elder person to watch people fishing.

- Ask the students to visit fish market in their area and get information of various fish by observing them.

2. Area: Clothing

2.1 Cloth Production

Unit-1 : Actually handling a boll/sliver of cotton and a spindle

This is Spindle speaking...

Hey, kids, do you see me? I'm Spindle. How do you do?

But, we don't know what a Spindle looks like.

Well, that's just why I am introducing myself. So, here I am …. Spindle.

You know that cloth is mode from cotton. But first the cotton has to be drawn into a yarn or thread. This is called spinning. The simplest device for spinning yarn is me ….. Spindle.

Now, what did you say? That machines are used for spinning. You are right, there. But before the machine was invented, I had to help produce the yarn. Then the spinning wheel was invented. The spindle and the wheel were used to draw yarn in every household. And cloth was woven on a manual loom or handloom. This work is carried on in some households even today.

Children, I am not a machine. My structure is very simple. I have only two parts, a shaft and a whorl.

My shaft is made of steel. It is about as thick as a thick incense stick *(agarbatti)* and is 18 cm long. The top end of the shaft is flattened and has a notch. About 2.5 cm from its lower end a brass disc or 'whorl'

is fitted to the shaft. My lower end is sharp and is called the tip or point.

So, this is me ……. a Spindle

What did you say? Where's the cotton required to make the thread? Well, that's mighty clever of you! Last year you learnt to clean the cotton didn't you? That clean cotton is carded and shaped into a thick, full wick-like shape about 15 to 18 cm long. This is the 'sliver' from which the yarn is drawn.

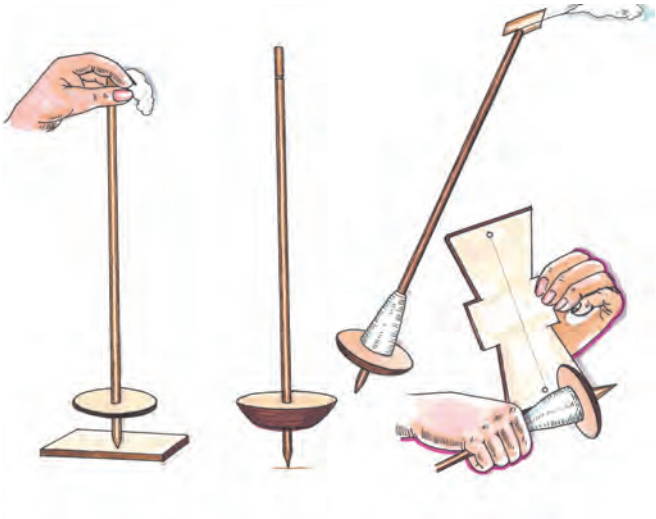
The 'sliver' and I, spindle, together spin the yarn. What's that? You'd like to touch me? Sure, you're welcome!

As you do so, do measure my length and the size of the whorl. See how full is the sliver of cotton. And, yes, do take care that my sharp tip or point does not poke anyone.

You can try turning me around.

But we shall learn to draw the yarn only next year, that is, in Std IV.

Sliver



My Activity

Learn by heart :

(1) See how the spindle goes
Spin, spin, spin.
Drawing yarn from the cotton
Long and thin.

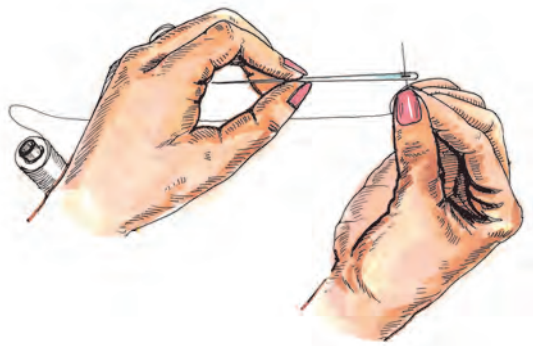
(2) Make a model of a spindle using a long, round, thick stick and a piece of cardboard.

- Show an actual spindle or a picture, for the children to observe.
- Give them information about spinning.

2.2 Basic Tailoring

Threading a Needle

Let's play a game.



How does the game help?

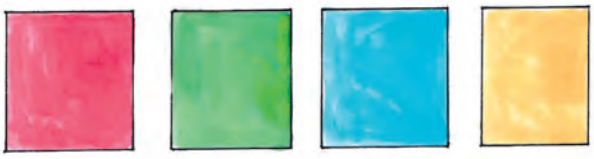
1. Eyes become sharper.
2. We develop hand-eye coordination.
3. We can do basic stitching.
4. We learn to work with each other's help.

Take care while playing :

1. Do not poke each other with the needles.
2. When the game is over place the threaded needles in a small box or pin it to a paper and put it away carefully.

My Activity

1. Thread a needle once every day.
2. Collect samples of cloth from some one who does tailoring and stick them in a scrap book.



◆ Give information about tailoring.

2.3 Doll Making

You like to wear a mask, don't you?

We have a lot of fun wearing a mask on occasions like a birthday, *Holi* or *Rangapanchami*. We can buy masks but let us learn to make one.

Materials : A big paper plate (about the size of your head), a thick needle and thread, scissors, red and black felt pens, glue, plain yellow and plain white paper, pictures of animals, birds, flowers, fruits, etc.

Do this:

1. Use the black and red sketch pens appropriately to draw eyes, nose, ears, lips, etc. on the outer surface of the paper plate.
2. Draw a picture of an animal, bird, flower, fruit that you like. Cut out the shape of the eyes from the picture.
3. Make holes on the two sides of the plate near the ears and pass a thread through each.
4. The threads should be long enough to go around the head and be tied into a knot.
5. Place the mask on your face and tie the knot behind your head.

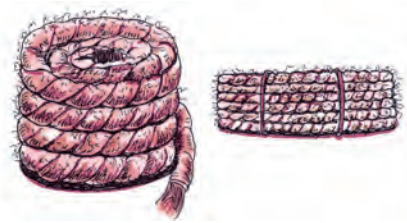
My Activity

Make masks of different shapes.

- Demonstrate how to make the mask, in the classroom, or show a picture of one and tell children to make masks.
- Pictures on this page are of masks that children have themselves made.

2.4 Coir Work

Write the names and uses of articles made from coir or coconut.



What do you see in the picture?

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Uses

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What do you see in the picture?

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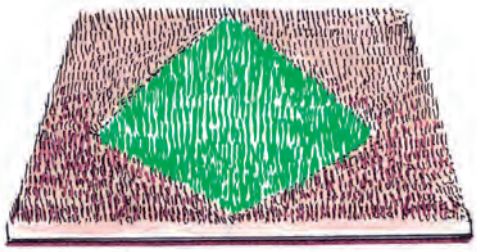

What do you see in the picture?

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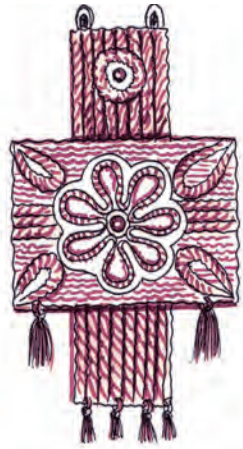

What do you see in the picture?

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What do you see in the picture?

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What do you see in the picture?

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My Activity

(1) Write the uses of things as shown in the picture.

(2) Draw a picture of one thing in your house that is made from coir.

- Show the class the things that are available. If possible arrange a visit to a coir centre.

3. Area : Shelter

3.1 Clay Modelling

Making the shapes of fruits, vegetables, etc. in clay.

Materials : Clay, water, a wooden board, a pointed object, a large pan, a broom, duster, etc. for cleaning.

Method

1. Guava :

- Observe a guava. Take a lump of clay.
- Give it a slightly tapering shape at one place.
- Make a small depression one side for the stalk. Make a stalk and stick it there and shape it neatly. With a wet hand, smoothen the shape.

2. Apple :

- Observe an apple. Make a round ball of clay.
- To give the dipped shape, press with your thumb on opposite sides. Make the stalk and stick it there.

3. Bottle Gourd :

- Observe a bottle gourd. Make a longish shape from clay. Shape it like a bottle gourd and fix a stalk on it.

My Activity

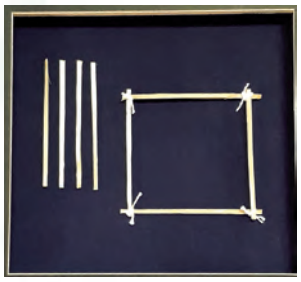
Observe fruits and vegetables and make clay fruits and vegetables like them.

- Show those things in the class that are actually available. Explain to the students the details of clay work. If possible, arrange a field visit.

3.2 Bamboo and Cane Work

Making different shapes using bamboo strips and sticks.

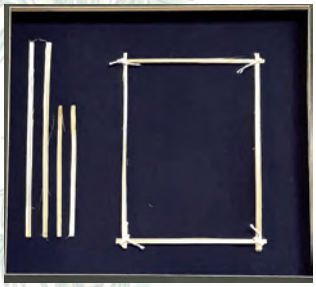
Materials and tools : Four bamboo strips of equal lengths (for example, all 15 cm long or all 10 cm long). Strong thread.



Procedure

1. Take four bamboo strips of equal length.
2. Place their ends on each other as shown in the figure.
3. Tie them together firmly with a thread.

Can you tell what shape this is ?



1. Take two bamboo strips of equal length.
2. Take another two strips of equal length but different in length from the first two.
3. Place the sticks as shown in the figure and tie the ends together firmly with a thread.

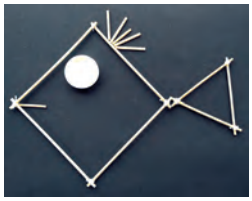
Can you tell what shape this is ?

1. Take three bamboo strips of equal length.
2. Place their ends on each other as shown in the figure.
3. Tie them tightly with a thread.

Which shape have you made now?

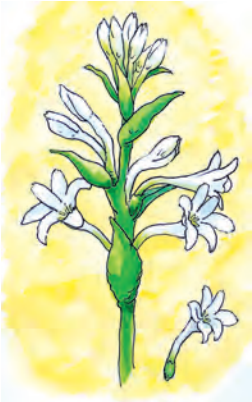
My Activity

Take bamboo strips of different lengths and make a variety of shapes.



- In view of the age of the children avoid using tools with sharp points or edges. Give guidance where necessary.

3.3 Cultivation of Flowering and Ornamental Plants.



Finding out which flowers and buds are used to make garlands, *gajaras*, etc.

Gajara - A small garland, 15 to 30 cms made from small flowers and their buds, is called a *gajara*. The flowers may be threaded together or their stalks may be woven together to make a *gajara*. *Gajaras* are worn in their hair by women and girls.

Mogara, jaai, jui, aboli, chameli, bakul, tuberose are the flowers used to make *gajaras*. Buds of flowers like the *mogara*, *jaai, jui, chameli* are plucked and kept in the water along with their stalks when they are ready to open. These flowers remain fresh for a longer time.

Large compact flowers / Garlands - (i.e. flowers whose many petals grow close to each other) and mango, *ashoka* or *tagar* leaves are threaded artistically on a string to make long garlands. Such garlands are used on occasions like religious functions, welcome or farewell functions, for statues, wedding ceremonies, etc.

Marigolds, chrysanthemums, tuberose, *mogara*, asters, *tagar, aboli* are used for these garlands. These garlands are thick and long.

A bouquet of flowers -

Some of the flowers used for garlands can also be used to make a bunch or bouquet of flowers for decorations or as a gift. These flowers can also be arranged artistically for use during a function.

My Activity

1. Go to a flower shop and observe the *gajaras*, garlands, *torans* displayed there. 2. Make a small garland.

Other Areas

Conservation of Animals and Birds.

Unit : To study how cows, buffaloes, goats, fowl are useful to humans.

Characters : Children wearing masks of a cow, buffalo, goat and hen.

Stage : Make place in the centre of the classroom.

Dialogue

A hen picking grain (A girl in the role of a hen acts the part.)

Cow (Enter) : Hey, Mrs Hen! Haven't you finished digging up the yard?

Hen : Oh dear, you've come so early. Where did the cowherd go?

Cow : Do you know what happened today! The Master and Mistress both went to the neighbouring village early this morning. And they won't be coming back till late in the evening. So, we told the cowherd, "Why don't you also take a day off today?" He promptly ran away. So, we too came back soon!

Hen : But, do you know why Master and Mistress have gone? They're going to sell all those eggs we gave and bring home a bagful of money!

Cow : Oh well, you need not boast, you know. We are also very useful. Our milk too fetches a lot of money. Besides, yoghurt, butter, *ghee, khoya, paneer* also sell well and are a good source of income.

Buffalo (Enter): Oh Tambutai! Don't show off so much! We give much more milk than you do. And it doesn't spoil easily. So many products are

made from our milk and there's a great demand for it, too.

(Goat comes hopping in.)

Hen : Here's our wild friend Goat ! So where have you been gallivanting today?

Goat : You know Sister Hen, we don't stay at home pecking at the ground like you. We go and find our own food. No one has to spend money for us. And yet we have our uses. We give milk and meat and all our parts as well as our droppings are useful, you know.

Cow : That's true of course, but after all we are the ones who provide bullocks for tilling the fields and cows for milk.

Buffalo : Oh, you two can keeping talking but neither of you can compare with us.

(Looking this side and that) Now, Oh dear! Where's Hen disappeared. She must be foraging in the garbage.

Hen : (*Coming up)* I'm right here. I noticed a kite hovering in the sky. It makes off with my chicks. So I shooed them into the pen. But I was listening to all that you said. The long and short of it is that we are all useful to humans. Why should we quarrel over that?

Buffalo : She's right. She looks small but she makes great sense. Anyway, the sun is setting and they'll soon be back. Let's all go back to our places. *(All leave.)*

My Activity

1. Enact this dialogue in a class or in a school function.
2. Collect pictures of cows, buffaloes, goats and hens.

- Which are the domestic animals ? Explain the uses of domestic animals. Arrange a field visit.
- The dialogue given above is only suggestive. Suitable changes can be made in it as needed.